

re

```
In [1]: import re
```

```
In [2]: txt = "Hi, how are you today?"  
re.search('Hi', txt)
```

```
Out[2]: <re.Match object; span=(0, 2), match='Hi'>
```

```
In [4]: txt.find('Hi')
```

```
Out[4]: 0
```

```
In [5]: s = re.search('Hi', txt)  
s != None
```

```
Out[5]: True
```

Identifiers

```
In [6]: # . = any character, except the new line  
print(re.search('a.', 'aaaaa') != None)  
print(re.search('a.', 'aa') != None)  
print(re.search('a.', 'ab') != None)  
print(re.search('a.', 'a1') != None)  
print(re.search('a.', 'a#') != None)  
print(re.search('a.', 'a') != None)  
print(re.search('a.', 'a\n') != None)
```

```
True  
True  
True  
True  
True  
True  
False  
False
```

```
In [7]: # ? = match 0 or 1 repetitions  
print(re.search('ba?b', 'bb') != None)  
print(re.search('ba?b', 'bab') != None)  
print(re.search('ba?b', 'abab') != None)  
print(re.search('ba?b', 'baab') != None)
```

```
True  
True  
True  
False
```

```
In [8]: # * = match 0 or more repetitions  
print(re.search('ba*b', 'bb') != None)
```

```
print(re.search('ba*b', 'bab') != None)
print(re.search('ba*b', 'baab') != None)
print(re.search('ba*b', 'baaaaaaab') != None)
print(re.search('ba*b', 'baaaaaagb') != None)
```

True
True
True
True
False

```
In [9]: # + = match 1 or more repetitions
print(re.search('ba+b', 'bb') != None)
print(re.search('ba+b', 'bab') != None)
print(re.search('ba+b', 'baab') != None)
print(re.search('ba+b', 'baaaaab') != None)
```

False
True
True
True

```
In [10]: # {} = matches a range of number of times
print(re.search('ba{1,3}b', 'bab') != None)
print(re.search('ba{1,3}b', 'baab') != None)
print(re.search('ba{1,3}b', 'baaab') != None)
print(re.search('ba{1,3}b', 'baaaaab') != None)
print(re.search('ba{1,3}b', 'bb') != None)
```

True
True
True
False
False

```
In [11]: # ^ = matches start of a string
print(re.search('^a', 'ab') != None)
print(re.search('^a', 'abcde') != None)
print(re.search('^a', ' ab') != None)
```

True
True
False

```
In [12]: # $ = matches end of a string
print(re.search('a$', 'ba') != None)
print(re.search('a$', 'bcdea') != None)
print(re.search('a$', 'badddd') != None)
```

True
True
False

```
In [13]: # Example-1
print(re.search('^a.a$', 'aba') != None)
print(re.search('^a.a$', 'a a') != None)
print(re.search('^a.a$', 'abbba') != None)
```

True
True
False

```
In [14]: # Example-2
print(re.search('^a.?a$', 'aba')!=None)
print(re.search('^a.?a$', 'aa')!=None)
```

True
True

```
In [15]: # Example-3
print(re.search('^a.*a$', 'aba')!=None)
print(re.search('^a.*a$', 'a          a')!=None)
print(re.search('^a.*a$', 'abcdefgha')!=None)
```

True
True
True

```
In [16]: # Example-4
print(re.search('^a.{1,3}a$', 'aba')!=None)
print(re.search('^a.{1,3}a$', 'abcda')!=None)
print(re.search('^a.{1,3}a$', 'abcdefa')!=None)
```

True
True
False

```
In [ ]:
```

```
In [17]: # [] are used to specify a set of characters to match
print(re.search('[123abc]', 'hi')!=None)
print(re.search('[123abc]', 'welcome')!=None)
print(re.search('[123abc]', '1abc')!=None)
print(re.search('[123abc]', 'hi2')!=None)
print(re.search('[1-3a-c]', 'welcome')!=None)
```

False
True
True
True
True

```
In [18]: # () are used to determine a group of characters (together)
print(re.search('(abc)', 'abc')!=None)
print(re.search('(abc){2,3}', 'abc')!=None)
print(re.search('(abc){2,3}', 'abcaabc')!=None)
print(re.search('(abc){2,3}', 'abcabcaabc')!=None)
print(re.search('(abc){2}', 'abcabcaabc')!=None)
print(re.search('^ (abc){2,3}$', 'abc')!=None)
print(re.search('^ (abc){2,3}$', 'abcaabc')!=None)
print(re.search('^ (abc){2,3}$', 'abcabcaabc')!=None)
```

True
False
True
True
True
False
True
True

```
In [19]: # | vertical bar is a logical or
print(re.search('abc|123', 'a') != None)
print(re.search('abc|123', '1') != None)
print(re.search('abc|123', '123') != None)
print(re.search('abc|123', 'abc') != None)
```

False
False
True
True

```
In [21]: # \ backslash is used with special characters
print(re.search('a?', '123') != None) # search for a 0 or 1 time
print(re.search('a\\?', '123') != None) # search for a?
```

True
False

```
In [23]: # Special Characters
# \d = matches any decimal digit [0-9]
print(re.search('\\d', 'welcome123') != None)
print(re.search('\\d', '123') != None)
print(re.search('\\d', 'welcome') != None)
```

True
True
False

```
In [24]: # \D = matches any non-digit character
print(re.search('\\D', 'welcome123') != None)
print(re.search('\\D', 'welcome') != None)
print(re.search('\\D', '123') != None)
```

True
True
False

```
In [25]: # \w = matches any alphanumeric character
print(re.search('\\w', 'welcome123') != None)
print(re.search('\\w', 'welcome') != None)
print(re.search('\\w', '123') != None)
print(re.search('\\w', 'welcome123.,?') != None)
print(re.search('\\w', '!.?#') != None)
```

True
True
True
True
False

```
In [26]: # \\W = matches any non-alphanumeric character
print(re.search('\\W','welcome123')!=None)
print(re.search('\\W','welcome')!=None)
print(re.search('\\W','123')!=None)
print(re.search('\\W','welcome123.,?')!=None)
print(re.search('\\W','.?#')!=None)
```

False
False
False
True
True

```
In [27]: # \\s = matches any white space character
print(re.search('\\s','welcome every one')!=None)
print(re.search('\\s','welcome')!=None)
print(re.search('s','welcome\\teveryone')!=None)
```

True
False
False

```
In [28]: # \\S = matches any non-white space character
print(re.search('\\S','welcome every one')!=None)
print(re.search('\\S','welcome')!=None)
print(re.search('\\S','welcome\\teveryone')!=None)
print(re.search('\\S',' \\t\\n')!=None)
```

True
True
True
False

```
In [ ]: # \\t = tab
        # \\n = new line
        # \\.= the character .
```

Some functions

```
In [ ]: # re.split('expr', txt)
        # re.findall('expr', txt)
```

```
In [42]: txt='''welcome everyone.
Today is Thursday, and we are so happy.
Next week we have the spring break.
My current gpa is 3.5
I hope to make it 4 soon\
my email is ha2285@nyu.edu
and my alternative email is: hasan.aljabbouli@nyu.edu
and my personal email is: abc@gmail.com'''
# txt.split(" ")
re.split("\\W",txt)
re.findall("\\d",txt)
re.findall("\\D*", txt)
```

```
re.findall("[a-z]+",txt)
re.findall("\\w+@\\w+\\.\\w+", txt)
```

Out[42]: ['ha2285@nyu.edu', 'aljabbouli@nyu.edu', 'abc@gmail.com']

```
In [ ]: # txt.split('\n')
# re.split(' +', txt)
# re.split('\s+', txt)
# re.findall('[a-zA-Z]+', txt)
```